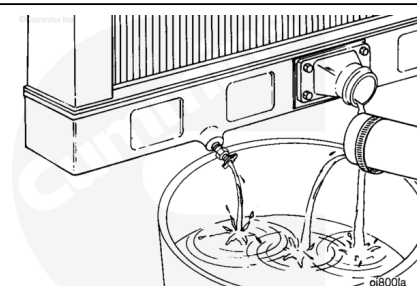


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining coolant that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

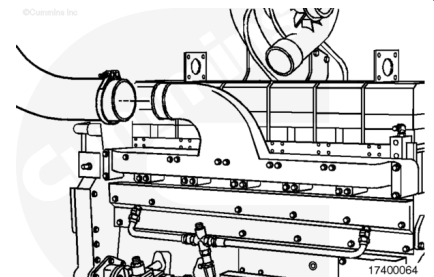
- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).
- Remove the air crossover pipe. Refer to Procedure 010-019 (</qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html>).
- Remove the aftercooler vent line. Refer to Procedure 008-017 (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>).
- Remove the aftercooler coolant inlet and outlet hoses. Refer to Procedure 010-006 (</qs3/pubsys2/xml/en/procedures/20/20-010-006.html>).
- Remove the fuel tubing mounting clamp and fuel tubes. Refer to Procedures 006-013 (</qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html>) or 006-024

(/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html).

- Remove the wiring harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/13/13-019-043.html) in the Troubleshooting and Repair Manual Electronic Control System QSK19 CM850, Modular Common Rail System Series Engine, Bulletin 4021493.
- Remove the intake manifold pressure sensors. Refer to Procedure 019-061 (/qs3/pubsys2/xml/en/procedures/13/13-019-061.html) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail, Bulletin 4021493.
- Remove the intake manifold temperature sensors. Refer to Procedure 019-059 (/qs3/pubsys2/xml/en/procedures/13/13-019-059.html) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail, Bulletin 4021493.

Rail Applications

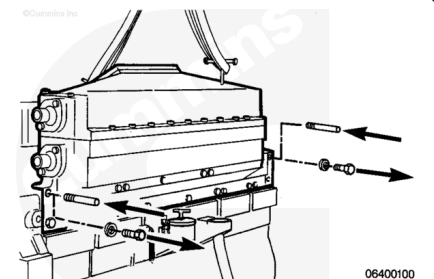
- Remove the turbocharger drain line from the scavenge cross and the turbocharger.
- Loosen the hose clamp and remove the hose.



Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



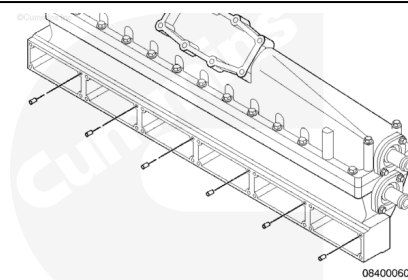
Remove two mounting capscrews. Install two 9.525 mm [0.375 in] guide studs.

Use a hoist, two tee handles, and a lifting sling. Attach the tee handles, sling, and hoist. Raise the hoist until there is tension on the sling.

Remove the remaining capscrews and the aftercooler assembly.

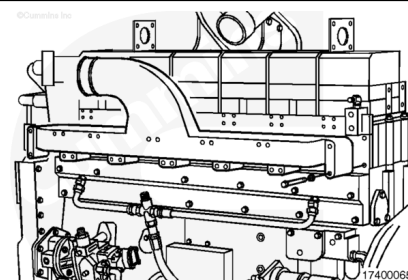
Remove and discard the gaskets.

Remove and discard the aftercooler bolt seals from the aftercooler housing.



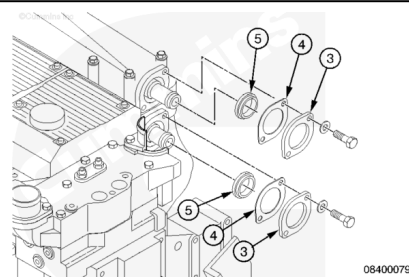
Rail Applications

Remove the 24 capscrews and intake manifold.

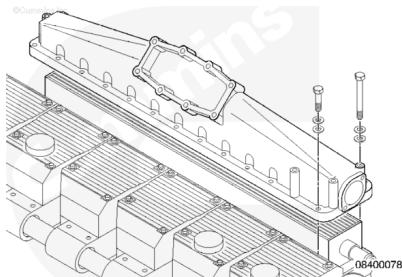


Disassemble

Remove the capscrews, seal retainer (3), gaskets (4), and seals (5). Discard the seals and gaskets.



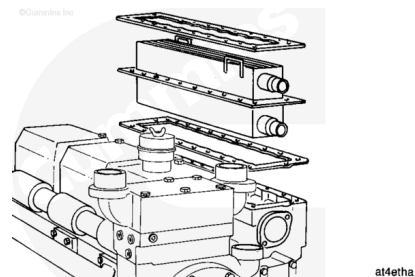
Remove the aftercooler cover capscrews and cover.



Remove the element and gaskets.

Inspect the cover and intake manifold gasket surfaces for damage.

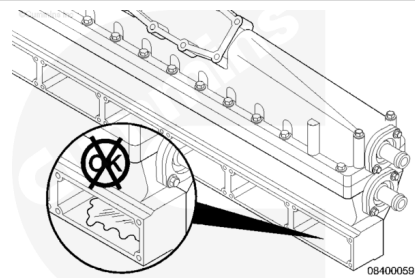
Discard the gaskets.



Inspect for Reuse

Clean and inspect the aftercooler for reuse. Look into the cylinder head ports and into the aftercooler for indications of coolant leakage. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>).

Replace the aftercooler core if leaks are found.

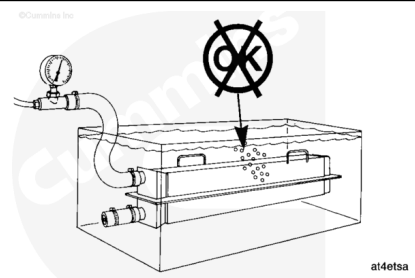


Use a hose plugged with a pipe plug. Plug one water tube as shown.

Attach an air line and gauge to the remaining tube.

Pressure check the element.

Apply air pressure and check for leaks.

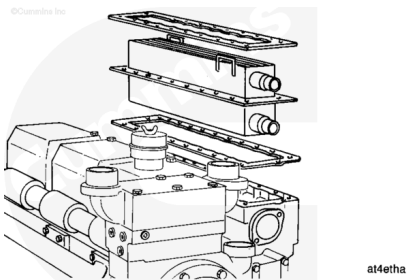


kpa		psi
415	MAX	60

Note : Heating the tank to 50°C [120°F] will improve the test results.

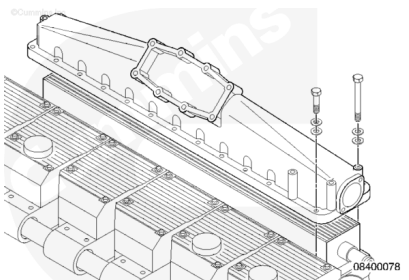
Assemble

Install the element and two new gaskets.



Install the aftercooler cover and capscrews.

Do **not** tighten the capscrews until after the seals are installed.



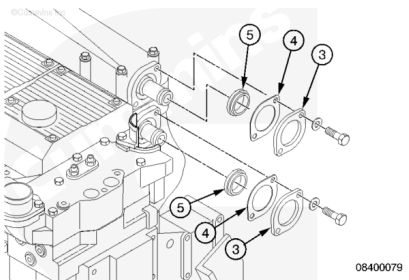
Use clean engine oil to lubricate the seals.

Install the seal on the retainer.

Use clean engine oil to lubricate the outside dimension of the water bosses on the aftercooler element.

Slide the gasket on the water bosses. Slide the seal and retainer on the tube.

Install the capscrews and tighten.

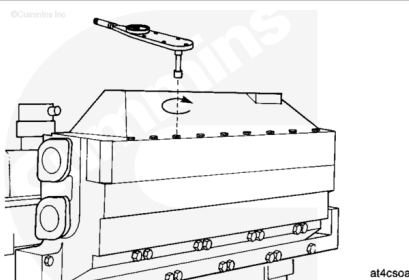


Torque Value: 35 n•m [25 ft-lb]

Install the 24 capscrews.

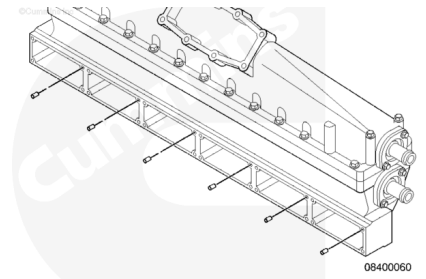
Tighten using the following steps and values.

Torque Value:	Step 1	35 n.m [25 ft-lb]
	Step 2	50 n.m [35 ft-lb]



Install

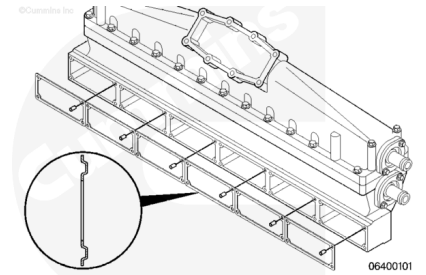
Install the bolt seals in the aftercooler housing.



Note : Use guide studs long enough to protrude beyond the installed part.

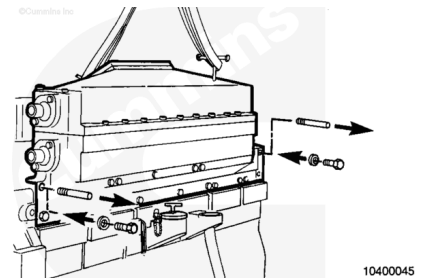
Install 12 guide studs in the upper row of capscrew holes in the cylinder heads.

Install the gaskets on the studs with the raised bead on the gasket toward the cylinder head.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



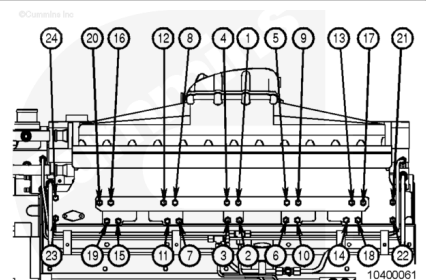
Install the aftercooler and the bottom row of capscrews.

Tighten the capscrews **only** enough to hold the part.

Remove the guide studs and install the top row of capscrews.

Use the sequence shown, and the steps and values indicated below to tighten the capscrews.

Torque Value:	Step 1	25 n.m [20 ft-lb]
	Step 2	45 n.m [35 ft-lb]

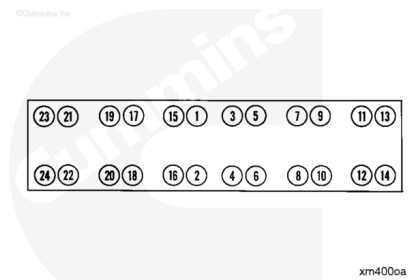


Rail Applications

Install the intake manifold and the 24 capscrews.

Use the sequence shown, and the steps and values indicated below to tighten the capscrews.

Torque Value:	Step 1	25 n.m [20 ft-lb]
	Step 2	45 n.m [35 ft-lb]



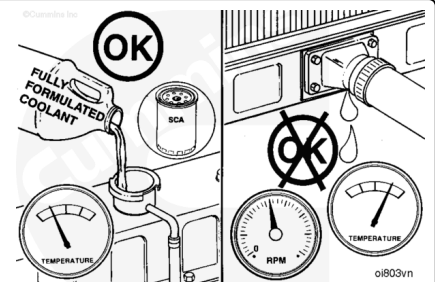
Finishing Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when filling the cooling system that the coolant is not spilled into the bilge area. Do not pump coolant overboard. If the coolant is spilled, it must be disposed of in accordance with local environmental regulations



- Install the intake manifold pressure sensors. Refer to Procedure 019-061 (</qs3/pubsys2/xml/en/procedures/13/13-019-061.html>) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail, Bulletin 4021493.
- Install the intake manifold temperature sensors. Refer to Procedure 019-059 (</qs3/pubsys2/xml/en/procedures/13/13-019-059.html>) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850 Modular Common Rail, Bulletin 4021493.
- Install the wiring harness. Refer to Procedure 019-043 (</qs3/pubsys2/xml/en/procedures/13/13-019-043.html>) in the Troubleshooting and Repair Manual, Electronic Control System, QSK19 CM850, Modular Common Rail, Bulletin 4021493
- Install the fuel tubing mounting clamp and fuel tubes. Refer to Procedures 006-013

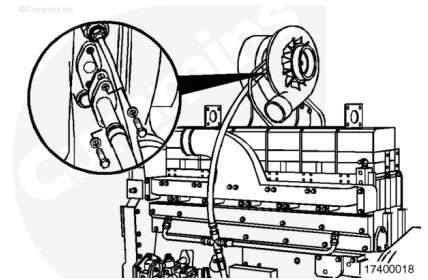
(/qs3/pubsys2/xml/en/procedures/20/20-006-013-tr.html) or 006-024

(/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html).

- Install the aftercooler coolant inlet and outlet hoses. Refer to Procedure 010-006 (/qs3/pubsys2/xml/en/procedures/20/20-010-006.html).
- Install the aftercooler vent line. Refer to Procedure 008-017 (/qs3/pubsys2/xml/en/procedures/20/20-008-017.html).
- Install the crossover pipe. Refer to Procedure 010-019 (/qs3/pubsys2/xml/en/procedures/20/20-010-019-tr.html).
- Fill the cooling system and check for leaks.

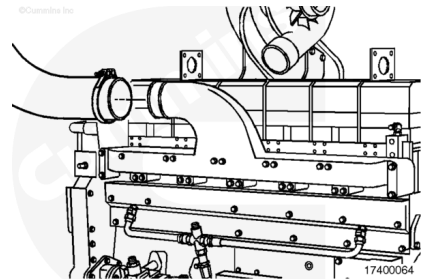
Rail Applications

Install the turbocharger oil drain line at the turbocharger.



Install the face seal o-ring and install the turbocharger oil drain line to the scavenge cross.

Torque Value: 80 n•m [60 ft-lb]



Install the air inlet hose.

Tighten the hose clamp.

Torque Value: 8 n•m [71 in-lb]

